

MATHDNN_1_4

September 9, 2021

```
[1]: import numpy as np
import matplotlib.pyplot as plt
import random as r

[2]: from IPython.display import set_matplotlib_formats
set_matplotlib_formats('pdf', 'svg')

[3]: np.seterr(invalid='ignore', over='ignore') # suppress warning caused by
→division by inf

[3]: {'divide': 'warn', 'invalid': 'warn', 'over': 'warn', 'under': 'ignore'}

[4]: def f(x):
    return 1/(1 + np.exp(3*(x-3))) * 10 * x**2 + 1 / (1 + np.exp(-3*(x-3))) *
→(0.5*(x-10)**2 + 50)

def fprime(x):
    return 1 / (1 + np.exp((-3)*(x-3))) * (x-10) + 1/(1 + np.exp(3*(x-3))) * 20
→* x + (3* np.exp(9))/(np.exp(9-1.5*x) + np.exp(1.5*x))**2 * ((0.5*(x-10)**2
→+ 50) - 10 * x**2)

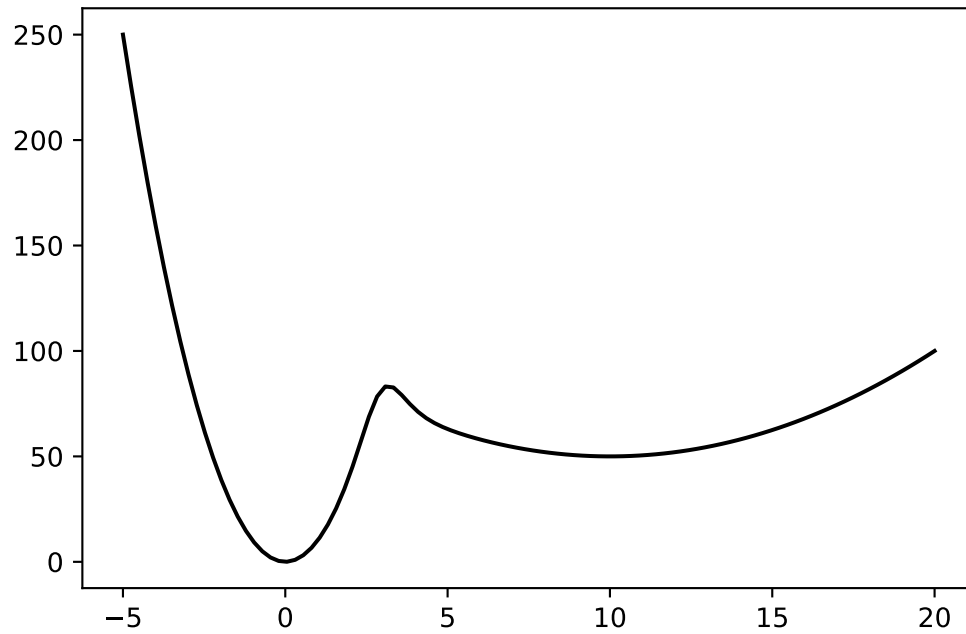
[5]: def GD_plot(alpha,run_time,num_of_sample):
    rt=[i for i in range(1,run_time+1)]
    for _ in range(num_of_sample):
        # randomly generating samples
        x=[r.uniform(-5,20)]
        for j in range(run_time-1):
            x.append(x[-1]-alpha*fprime(x[-1]))
        plt.plot(rt,x)
    plt.show()

[6]: def GD_freq(alpha,run_time,num_of_sample):

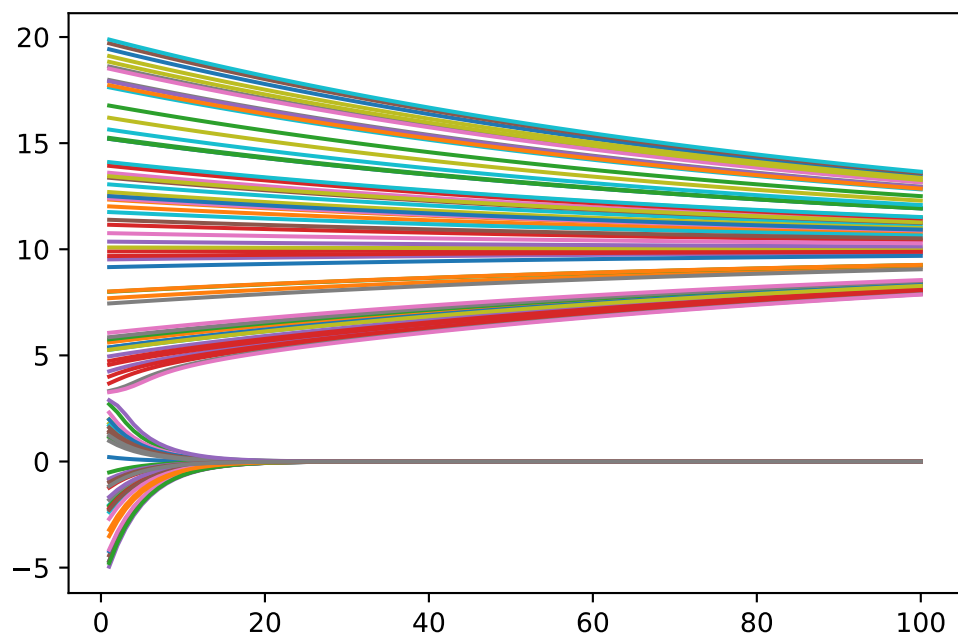
    num_0,num_10=0,0
    rt=[i for i in range(1,run_time+1)]
    for _ in range(num_of_sample):
        x=[r.uniform(-5,20)]
        for j in range(run_time-1):
            x.append(x[-1]-alpha*fprime(x[-1]))
        if 0.99*f(0)<=f(x[-1]) and f(x[-1])<=1.01*f(0):
```

```
    num_0+=1
    elif 0.99*f(10)<=f(x[-1]) and f(x[-1])<=1.01*f(10):
        num_10+=1
    return num_0/num_of_sample,num_10/num_of_sample
```

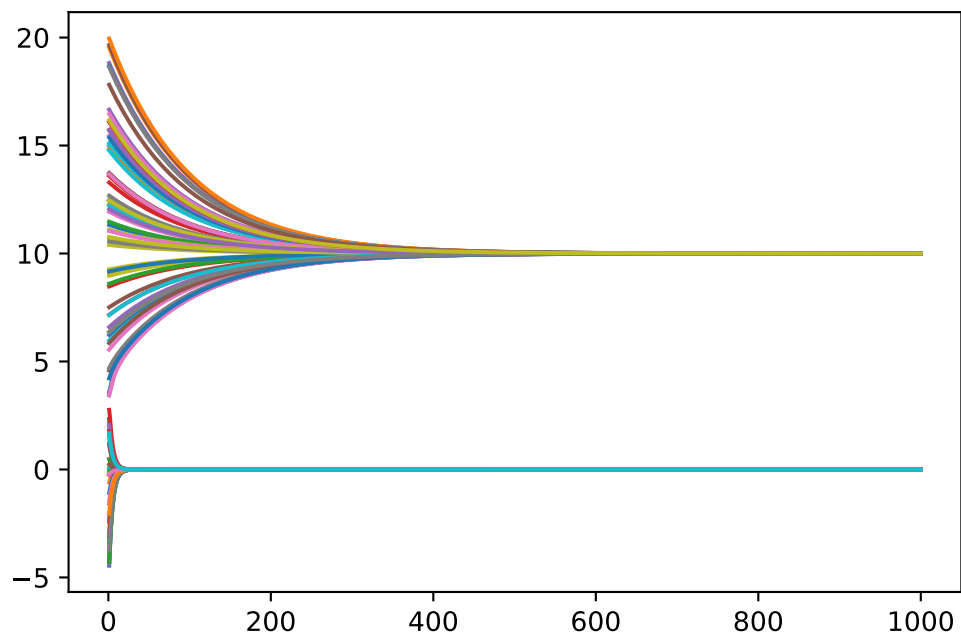
```
[7]: x = np.linspace(-5,20,100)
plt.plot(x,f(x), 'k')
plt.show()
```



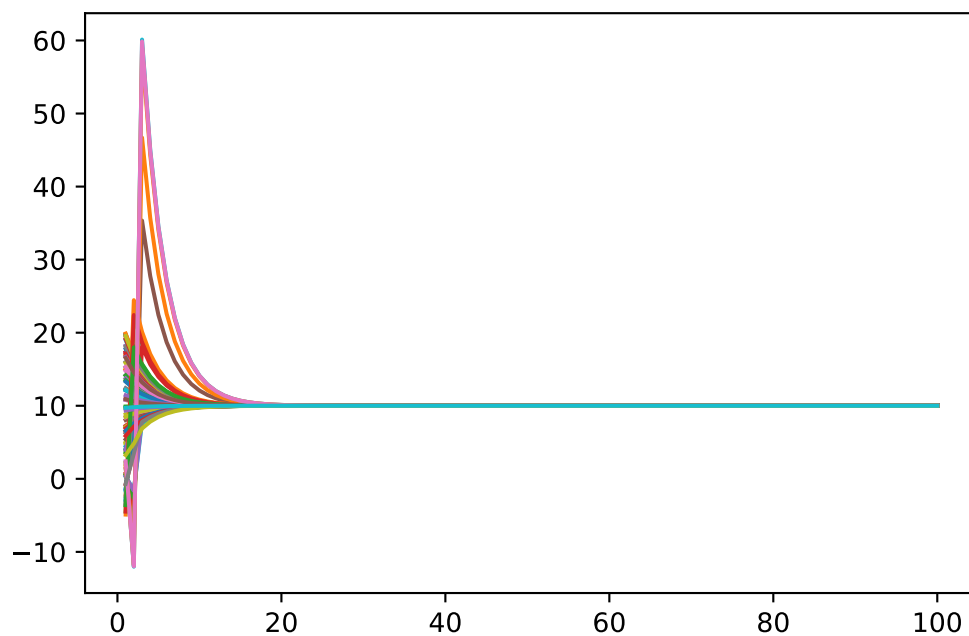
```
[8]: GD_plot(0.01,100,100)
```



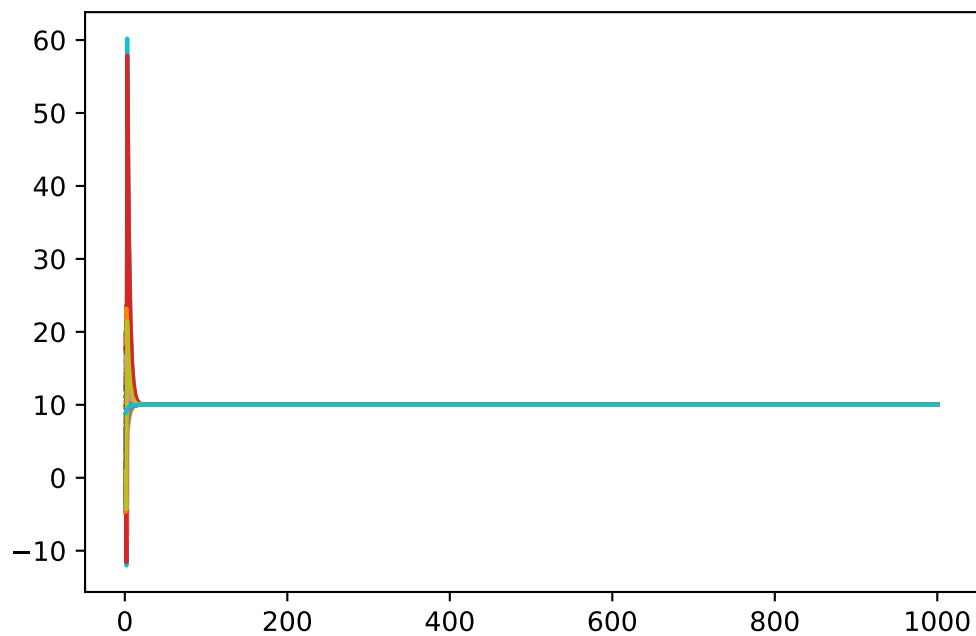
[9]: `GD_plot(0.01,1000,100)`



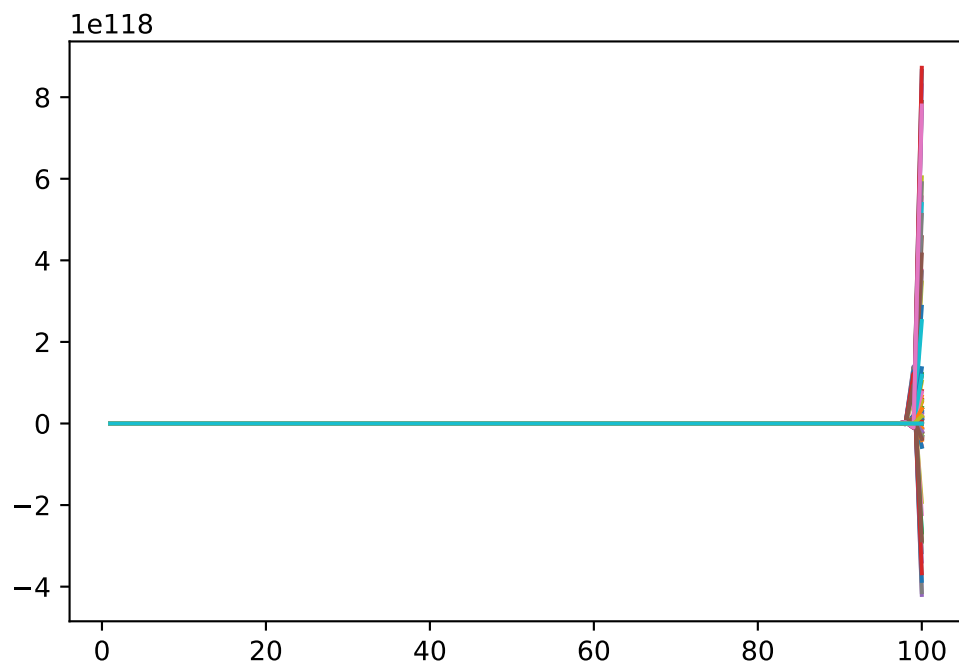
[10]: `GD_plot(0.3,100,100)`



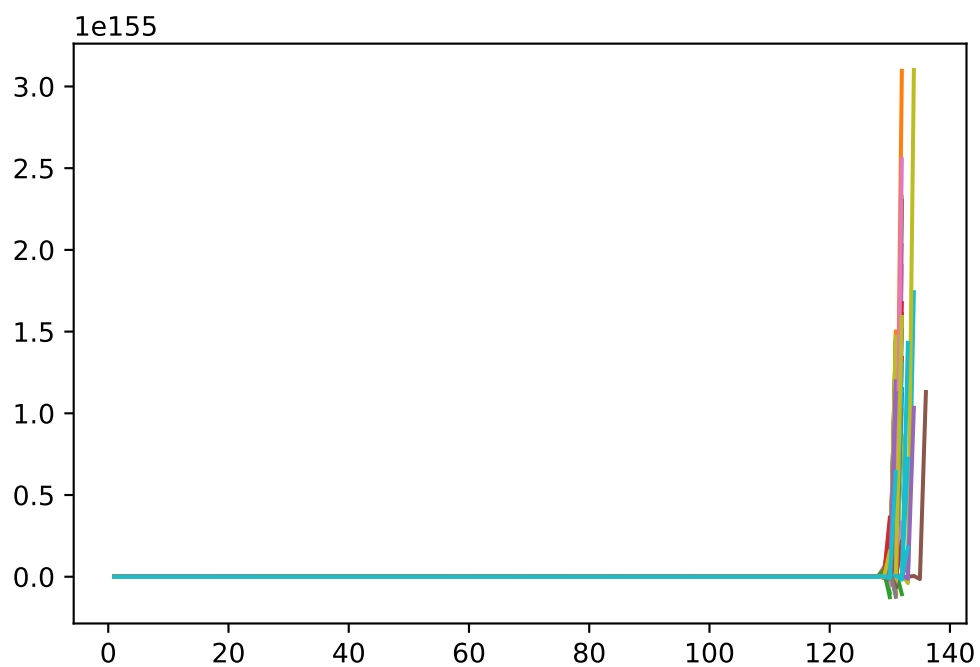
[11]: `GD_plot(0.3,1000,100)`



[12]: `GD_plot(4,100,100)`



[13]: `GD_plot(4,1000,100)`



```
[15]: #alpha=0.01
rt_1_to_0,rt_1_to_10=GD_freq(0.01,1000,300)
#alpha=0.3
rt_2_to_0,rt_2_to_10=GD_freq(0.3,1000,300)
#alpha=4
rt_3_div=1-sum(GD_freq(4,1000,300))
print(f"If rate is 0.01, approximately {round(100*rt_1_to_0,2)} percent of
→samples converge to 0 and {round(100*rt_1_to_10,2)} percent of samples
→converge to 10.")
print(f"If rate is 0.3, approximately {round(100*rt_2_to_0,2)} percent of
→samples converge to 0 and {round(100*rt_2_to_10,2)} percent of samples
→converge to 10.")
print(f"If rate is 4, approximately {round(100*rt_3_div,2)} percent of samples
→diverge.")
```

If rate is 0.01, approximately 30.33 percent of samples converge to 0 and 69.67 percent of samples converge to 10.

If rate is 0.3, approximately 0.0 percent of samples converge to 0 and 100.0 percent of samples converge to 10.

If rate is 4, approximately 100.0 percent of samples diverge.

```
[16]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
[ ]: !wget -nc https://raw.githubusercontent.com/brpy/colab-pdf/master/colab_pdf.py
from colab_pdf import colab_pdf
colab_pdf('MATHDNN_1_4.ipynb')
```

```
--2021-09-09 13:13:22-- https://raw.githubusercontent.com/brpy/colab-
pdf/master/colab_pdf.py
```

```
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
```

```
185.199.108.133, 185.199.109.133, 185.199.110.133, ...
```

```
Connecting to raw.githubusercontent.com
```

```
(raw.githubusercontent.com)|185.199.108.133|:443... connected.
```

```
HTTP request sent, awaiting response... 200 OK
```

```
Length: 1864 (1.8K) [text/plain]
```

```
Saving to: colab_pdf.py
```

```
colab_pdf.py          100%[=====>]    1.82K  --.-KB/s    in 0s
```

```
2021-09-09 13:13:22 (39.3 MB/s) - colab_pdf.py saved [1864/1864]
```

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

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